PGP_MITB_2610_IS613_SHAR LWIN KHIN

Singapore Management University

COURSE INFORMATION

Course Code & Title
IS613 - Cybersecurity Technology and Applications

Effective Start Date
17/08/2026

Offering Unit / Department / School
SCIS MITB

Course Career
Graduate - IT in Business

Course Units

CU:

1

Grading Basis
GRD - Graded

Course Description
This course provides an introduction to cybersecurity. The focus is on basic cryptographic techniques, user authentications, software security, and various network security topics. The course emphasizes on the applications of such technology in real-world business scenarios, with case studies that examine how these ideas can be used to protect existing and emerging applications. Examples include secure email communications, secure electronic transactions over the Internet, secure e-banking, data confidentiality and privacy in cloud computing, and secure protocols in realistic networking setups. Although the course covers fundamentals of cryptography, our emphasis is not on its mathematical background and security proofs, but rather on how such building blocks could be applied to satisfy business, communication, and networking needs.

Course Learning Outcomes

- Understand fundamental security concepts, models, algorithms, and protocols.
- Conduct basic software vulnerability and malware analyses and construct corresponding exploits.
- Design and implement secure user authentication on Internet facing servers.
- Formulate security requirements for real-world computing applications.
- Analyze latest security mechanisms in use.

SMU Graduate Learning Outcomes

Disciplinary knowledge, Multidisciplinary knowledge, Critical thinking & problem solving, Self-directed learning

Competencies

Critical Thinking, Business Needs Analysis, Programming and Coding, Digital Solutioning Skills

Course Area(s)

AREA - EngD Technical Application, RQCX - IS715, AREA - MITB Information Technology Management, AREA - MITB Cybersecurity Track Core

Simple Requirements

IS613/715 - Mutually Exclusive

Type

Antirequisite

Enroll in the following Courses:

- CS715 - Systems and Data Security

CLASS SECTION INFORMATION

Instructor(s)

SHAR LWIN KHIN - (lkshar@smu.edu.sg), YANG GUOMIN - (gmyang@smu.edu.sg)

Instruction Mode
In Person

Assessment Components (New)

Category*	Weightage*
Class Participation	10

Description
includes attendance 5%

Category*	Weightage*
Projects/Assignments	25

Description
Midterm

Category*	Weightage*
Projects/Assignments	5

Description
In-class quiz

Category*	Weightage*
Projects/Assignments	20

Description
Group project

Category*
Final Exam

Weightage*
40

Description
-

Instructional Methods and Expectations
N/A

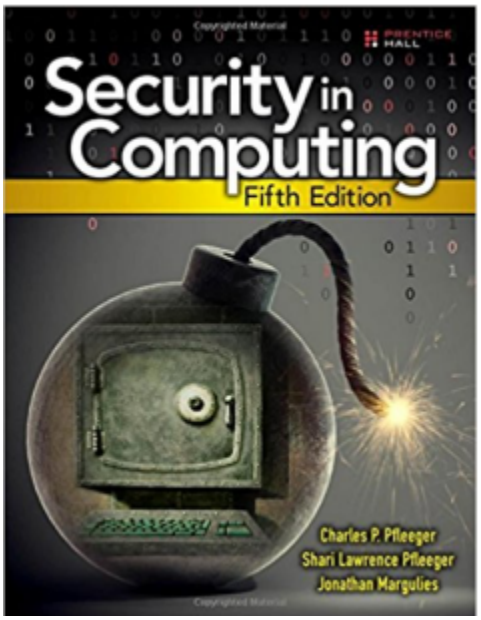
Weekly Lesson Plan

Session #	Lesson Activities	Student Assignment
1	Introduction	
2	Encryption basics & Symmetric encryption	
3	Asymmetric encryption	Quiz 1, Project team & topic submission
4	Message integrity and authentication	
5	Digital Certificates & SSL/TLS	
6	User authentication	Quiz 2
7	Midterm + Network security	Midterm (1:15 hr)
8	**BREAK**	

9	Software security	Quiz 3
10	Web security	Quiz 4, Project deliverables submission
11	Group project presentation	
12	**EXAM WEEK**	

Recommended Textbooks and Readings
Main Reading (articles, book chapters, videos, etc.):

Security in Computing (5th edition) by Pfleeger, Pfleeger, Margulies



Other Information
Resources

Tools:

Kali Linux

Metasploitable

Openssl

UNIVERSITY POLICIES

Academic Integrity

All acts of academic dishonesty (including, but not limited to, plagiarism, cheating, fabrication, facilitation of acts of academic dishonesty by others, unauthorized possession of exam questions, or tampering with the academic work of other students) are serious offences.

All work (whether oral or written) submitted for purposes of assessment must be the student's own work. Penalties for violation of the policy range from zero marks for the component assessment to expulsion, depending on the nature of the offence.

When in doubt, students should consult the instructors of the course. Details on the SMU Code of Academic Integrity may be accessed at https://smu.sharepoint.com/sites/oasis/SitePages/DOS-WKLSWC/UCSC.aspx.

Responsible Use of AI

The University recognises the importance of equipping students with the skills to navigate an AI-driven world. While AI tools may support learning, they must be used judiciously and responsibly. Students must not present AI-generated output as their own work.

Each course may set specific guidelines on the use of AI in assessments. Students are required to follow these guidelines carefully and should seek clarification from the instructors if uncertain.

Misuse of AI constitutes a breach of academic integrity and will be subject to the same penalties as other acts of academic dishonesty.

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Accessibility

SMU strives to make learning experiences accessible for all. Students who anticipate or experience physical or academic barriers due to disability are encouraged to inform the course instructor as soon as possible. Students may also contact the university's disability services team if they have questions or concerns about academic provisions: accessibility@smu.edu.sg. Please be aware that the accessible tables in our seminar room should remain available for students who require them.

Digital Readiness for Teaching and Learning (DRTL)

As part of emergency preparedness, instructors may conduct lessons online via the Zoom platform during the term, to prepare students for online learning. During an actual emergency, students will be notified to access the Zoom platform for their online lessons. The class schedule will mirror the current face-to-face class timetable unless otherwise stated.